

AZURE CA –Troubleshoot

During a campaign window, traffic spikes on a public-facing workload hosted on Azure Container Apps behind an Azure Application Gateway (WAF v2). Users report higher latency and sporadic timeouts. CPU on the only running revision instance is high, yet the service does not scale out. When scaling finally triggers, additional replicas intermittently fail to start. After the traffic peak, replicas do not scale back down for a long time, keeping costs elevated. In parallel, a worker Container App fails to start new replicas due to container image pull issues from Azure Container Registry (ACR). External access is inconsistent: Application Gateway reports backend health probe failures and receives fewer requests than expected.

This investigation unfolds in phases:

- scale-out not triggering
- scales but replicas fail to start (capacity/placement)
- never scales down
- image pulls from ACR fail
- external access broken due to Application Gateway / probe / HTTP settings.

1. Purpose of the task

- Make **scale-out** reliably trigger on correct signals (CPU/HTTP/queue) with sensible periods, statistics, and cooldown/stabilization windows.
- Ensure **real capacity** so new replicas start (ACA Environment/VNet integration, subnet/IP capacity, quotas).
- Restore timely and safe **scale-in** after the peak without flapping (min/max replicas, stabilization, connection draining).
- Fix **ACR image pull** via Managed Identity/RBAC and required egress/PE/DNS so rollouts succeed.
- Re-establish **stable public access** through correctly configured Application Gateway (listeners, probes, HTTP settings, hostnames, SG/NSG/UDR), with healthy backends and expected request volume.

2. Rules and restrictions

- **No application code changes and no image rebuilds.** Infra/config only.
- **Do not change the region or add new VNets.** Work within the provided RG/VNet/subnets.
- **Least privilege & minimal change:** every modification must be as small as possible and documented (what/where/why + evidence).

Allowed changes only in:

- **ACA scaling rules** (CPU %, HTTP RPS/requests in flight, Service Bus/Storage Queue backlog) + min/max replicas + stabilization window.
- **ACA revisions & service bindings** (without changing the image), load balancer health/probe path, timeouts.
- **ACA Environment / Networking:** VNet integration, subnet capacity (free IPs), egress (NAT Gateway) or Private Endpoints + Private DNS if ACR/logs are private.
- **ACR & Identity:** assign AcrPull to the Managed Identity (system- or user-assigned) used by Container Apps.
- **Application Gateway:** listeners (HTTP/HTTPS), HTTP settings (port, host header/SNI, timeout), health probes (path/interval/timeout), routing rules.
- **Monitoring:** Azure Monitor / Log Analytics configurations to verify outcomes (metrics, logs, autoscale history).
- **Validation required after each phase:** show metrics (CPU/RPS/backlog), autoscale history, probe/Backend Health status in Application Gateway, and a short load test (p95, error rate).
- **Do not delete existing resources;** fix configuration instead.

3. Best Practises and tips

<https://azure.microsoft.com/en-us/products/container-apps>

<https://learn.microsoft.com/en-us/azure/virtual-network/concepts-and-best-practices>

<https://learn.microsoft.com/en-us/azure/load-balancer/load-balancer-best-practices>

<https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-best-practices>

4. Additional requirements

Consider what modifications would be worth implementing to the current solution in terraform and general implemented infrastructure. Try to convince your teammate why it is worth changing the current approach in terms of, for example, costs, automation, optimization and maintenance, which are important from the perspective of a DevOps Engineer.

Additional Challenges

1. **Infrastructure as Code** – try to implement your own changed solution for upgraded infrastructure and terraform code.
2. **Diagram** – prepare diagram for current implemented infrastructure in order to visualize the concept

The above are required to complete the task, but if you have the time and motivation, you can add more functionality and best practices to your task 😊

Good luck!

Ten dokument i jego zawartość stanowią poufną własność właściciela kursu DevOps/Cloud, w którym uczestniczysz. Kopiowanie, dystrybucja lub publiczne ujawnianie – w całości lub w części – bez wyraźnej pisemnej zgody jest surowo zabronione. Wszelkie prawa zastrzeżone.